

Customer Shopping Behavior Analysis

End-to-End Data Analytics Project | Python • MySQL • SQL • Power BI

Professional Project Report

This report documents the analysis of **3,900 transaction records**. The workflow covers data quality assessment, exploratory analysis, SQL business analysis, and Power BI reporting.

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1. Executive Summary

The dataset contains **3,900 records** and generated total observed purchase revenue of **\$233,081.00**. The average purchase amount is **\$59.76**. After missing-value treatment, the average review rating is **3.75/5**.

The project demonstrates a complete analyst workflow: Python for preparation and EDA, MySQL/SQL for business questions, and Power BI for interactive reporting.

2. Business Questions

- Revenue by male vs. female customers.
- Discounted purchases above the overall average purchase amount.
- Top five products by average review rating.
- Standard vs. Express shipping spending.
- Subscriber vs. non-subscriber spending and revenue.
- Products with the highest discounted-purchase percentage.
- New, Returning and Loyal customer segmentation.
- Top three products within each category.
- Repeat buyers and subscription behavior.
- Revenue contribution by age group.

3. Data Quality & Preparation

Metric	Result
Rows	3,900
Original columns	18
Derived columns	purchase_frequency_days
Missing Review Rating	37
Median used for rating imputation	3.80
Duplicate rows detected	0
Total revenue	\$233,081.00
Average purchase	\$59.76

Column names were standardized to lowercase snake_case. Purchase-frequency categories were converted to a day-based analytical feature. The source contains 37 missing Review Rating values; these were treated using the median available rating.

4. Revenue by Gender

Gender	Purchases	Revenue	Avg Purchase
Female	1,248	\$75,191.00	\$60.25
Male	2,652	\$157,890.00	\$59.54

This view provides the revenue mix and average purchase value by gender.

5. Discounted Purchases Above Average

The overall average purchase is **\$59.76**. **839** discounted transactions exceed this benchmark. These records represent relatively high-value purchases made while a discount was applied.

6. Top Products by Review Rating

Product	Purchases	Average Rating
Gloves	140	3.86
Sandals	160	3.84
Boots	144	3.82
Hat	154	3.80
Skirt	158	3.79

7. Shipping Comparison

Shipping	Purchases	Avg Purchase	Revenue
Express	646	\$60.48	\$39,067.00
Standard	654	\$58.46	\$38,233.00

8. Subscription Analysis

Status	Purchases	Avg Spend	Revenue
No	2,847	\$59.87	\$170,436.00
Yes	1,053	\$59.49	\$62,645.00

9. Discount Penetration by Product

Product	Purchases	Discounted	Discount %
Hat	154	77	50.00%
Sneakers	145	72	49.66%
Coat	161	79	49.07%
Sweater	164	79	48.17%
Pants	171	81	47.37%

10. Customer Segmentation

New = 0 previous purchases; Returning = 1–5; Loyal = more than 5 previous purchases.

Segment	Customers
Loyal	3,476
Returning	424

11. Top Products by Category

Category	Product	Purchases	Rank
Accessories	Jewelry	171	1
Accessories	Belt	161	2
Accessories	Sunglasses	161	3
Clothing	Blouse	171	1

Category	Product	Purchases	Rank
Clothing	Pants	171	2
Clothing	Shirt	169	3
Footwear	Sandals	160	1
Footwear	Shoes	150	2
Footwear	Sneakers	145	3
Outerwear	Jacket	163	1
Outerwear	Coat	161	2

12. Repeat Buyers & Subscription

Customer Type	Customers	Subscribed	Subscription Rate
Non-Repeat Buyer	424	95	22.41%
Repeat Buyer	3,476	958	27.56%

13. Revenue Contribution by Age Group

Age Group	Purchases	Revenue	Contribution
18-24	486	\$29,258.00	12.55%
25-34	755	\$45,400.00	19.48%
35-44	729	\$43,463.00	18.65%
45-54	752	\$45,370.00	19.47%
55+	1,178	\$69,590.00	29.86%

14. SQL & MySQL

The cleaned data was loaded into the customer_shopping_analysis database and analyzed using SQL. The project uses aggregation, filtering, CASE expressions, subqueries, CTEs and window functions. Analysis views were prepared as a clean interface between SQL and Power BI.

15. Power BI Dashboard

Power BI was used as the presentation layer for KPIs and interactive business analysis, including revenue, customers, products, discounts, shipping, subscriptions and age-group performance.

16. Key Business Implications

- Use revenue and average-spend differences to improve customer targeting.
- Review high-value discounted purchases to understand whether promotions support meaningful spending.
- Combine product rating with purchase volume before prioritizing products.
- Evaluate shipping options using both average purchase and total revenue.
- Use customer segments to differentiate acquisition, retention and loyalty strategies.
- Monitor subscription adoption among repeat buyers as a potential retention indicator.

17. Limitations

The dataset is transaction-level and does not include transaction dates, product cost/margin, campaign identifiers or customer lifetime value. Therefore, the analysis describes observed purchasing behavior rather than profitability, time-series trends or causal effects. Customer IDs can repeat, so transaction counts should not automatically be interpreted as unique-customer counts.

18. Conclusion

This project demonstrates a complete end-to-end data analytics workflow: raw data preparation in Python, structured business analysis in MySQL/SQL, and interactive reporting in Power BI. It demonstrates practical skills in data cleaning, exploratory analysis, SQL, business intelligence, visualization and data storytelling.

Technology Stack

Python • Pandas • NumPy • MySQL • SQL • Power BI • GitHub